Ardeshir Aidun

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Professional Summary

Electrical Engineering student at UC Santa Barbara with hands-on experience in transistor-level CMOS design, FPGA development, embedded systems, and hardware debugging. Proficient in SystemVerilog, C++, Python, and SPICE, with a track record of designing, simulating, and validating hardware systems from concept through deployment.

Education

University of California, Santa Barbara
Bachelor of Science, Electrical Engineering. GPA: 3.72. — *Expected June 2028*
Relevant Coursework: Digital Design, Analog and Digital Circuits Fundamentals, Signal Analysis, Robotics: Control, Quantum Photonics, Research Process for Engineers

Projects

CMOS Standard Cell Library | SPICE, ngspice, Python — Aug. 2026 – Aug. 2026
- Designed and simulated a 5-cell CMOS standard cell library (inverter, NAND2, NOR2, XOR2, D flip-flop) at the transistor level using SPICE, sizing PMOS/NMOS devices and verifying functional correctness against truth tables
- Built a Python automation script to batch-run ngspice simulations and extract propagation delay (tpHL/tpLH) via threshold-crossing detection, replacing manual waveform analysis
- Identified and resolved two transistor-level logic bugs through waveform-level root-cause analysis: a transmission-gate wiring error in XOR2 and a clock-phase conflict in the D flip-flop feedback loop causing an unstable output node

1965 Thunderbird Taillight Controller | SystemVerilog, FPGA — Mar. 2026 – Mar. 2026
- Designed an 8-state Moore FSM in SystemVerilog to replicate sequential taillight behavior covering turn signal, brake, and hazard functions
- Implemented priority-encoded control logic to resolve input conflicts among simultaneous signals per a strict override specification
- Synthesized and deployed the design to an FPGA, verifying correct LED output across all input combinations against a truth table

Multi-Node ESP32 Mesh Network | C++, ESP32, ESP-NOW — June 2026 – July 2026
- Designed and built a 3-node wireless mesh network using ESP32 microcontrollers, with sensor nodes reporting live temperature and humidity data to a central gateway over ESP-NOW
- Architected a wireless OTA firmware update system using WiFi and HTTP, leveraging the ESP32's dual-partition bootloader to safely stage and verify updates without disrupting the running device

Ultrasonic Sensor Characterization | Python, Data Analysis — May 2026 – June 2026
- Designed and executed a bench test protocol to characterize accuracy, noise, and drift of an HC-SR04 sensor across a 5-30 cm range using an Arduino Uno
- Built a Python data pipeline to compute statistical metrics including mean error, standard deviation, and drift rate, visualized with matplotlib

Experience

Member — Oct. 2024 – Present
IEEE - UCSB Chapter — *Goleta, CA*
- Developed soldering and PCB design skills through hands-on technical workshops
- Assembled and troubleshot circuits, building practical proficiency in component assembly and prototyping

Guest Advocate — June 2025 – June 2026
Target — *San Jose, CA*
- Coordinated order pickups and curbside deliveries, resolving fulfillment and self-checkout issues

Leader — Aug. 2022 – May 2024
Link Crew — *Los Gatos, CA*
- Mentored incoming students through orientation and campus acclimation activities

Skills

Hardware and HDL: SystemVerilog, FPGA, Microcontrollers, Circuit Design, PCB Design, Oscilloscope, Bench Testing, Soldering, Prototyping, LTspice

Software and Programming: C++, Python, SPICE, ngspice, MATLAB, Git/GitHub, Technical Documentation

Communication Protocols: UART, ESP-NOW, WiFi/HTTP

Technical Concepts: Finite State Machines, Digital Design, Transistor-Level CMOS, Signal Analysis, Boolean Algebra